TECHNISCHE UNIVERSITÄT DRESDEN

DRESDEN concept

Research Project

Comparison of 2D and 3D Implementation of a Streamlined Body in Eddy-Dominant Flow

Institut für Wasserbau und Technische Hydromechanik, Professur Wasserbau

Presented by:

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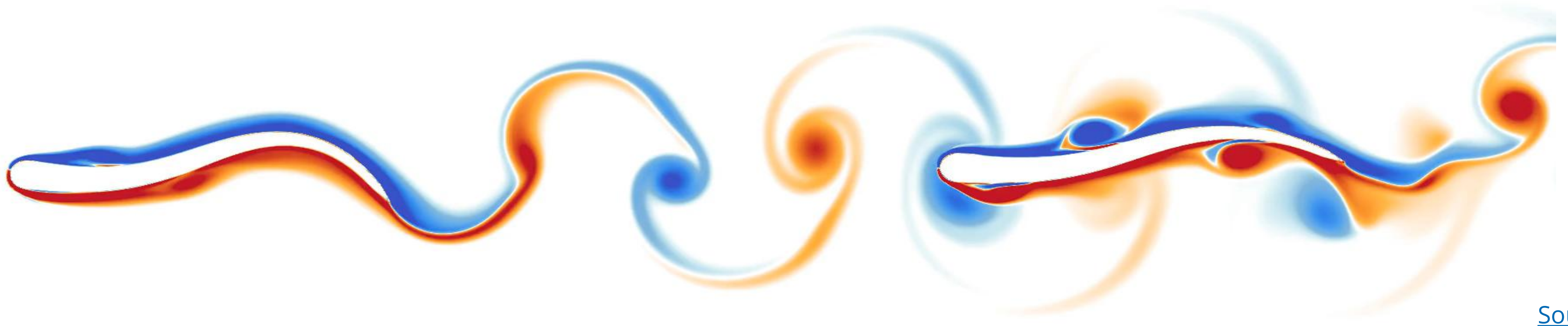
Motivation

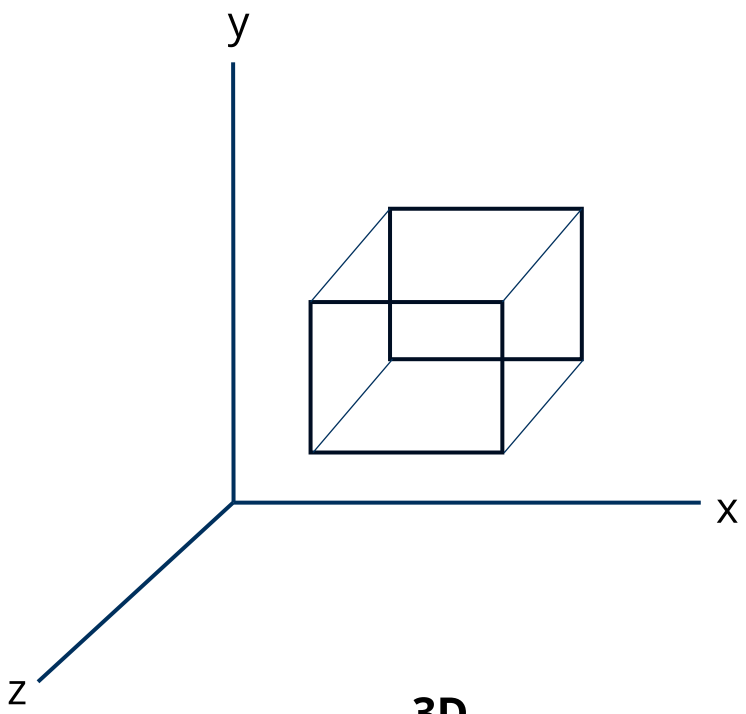
- Aquatic fauna face various disturbances.
- Important to know the flow characteristics and to predict them.
- Project aims to establish a comparison between 2D, and 3D model.
- Numerical Simulation employed for comparison.



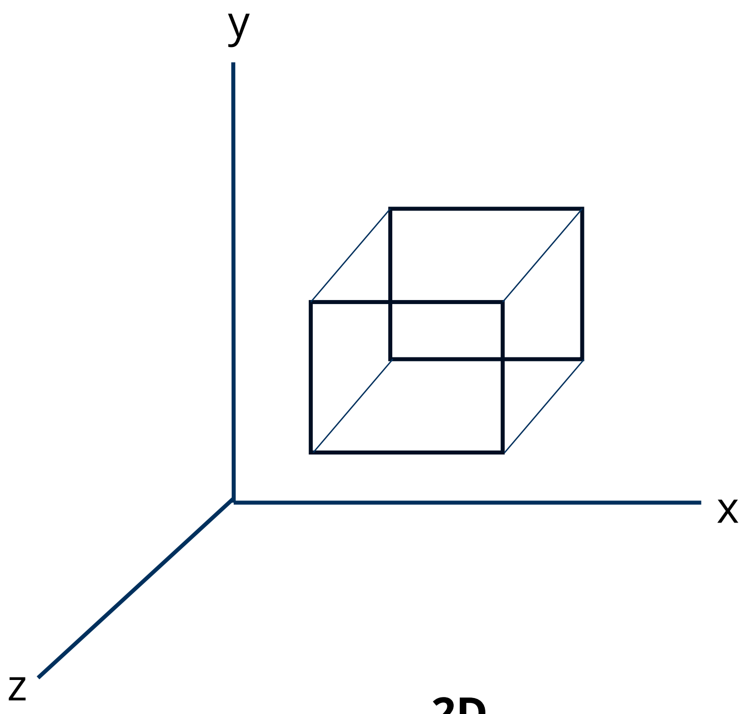
Source

Motivation

- Model simplification



3D

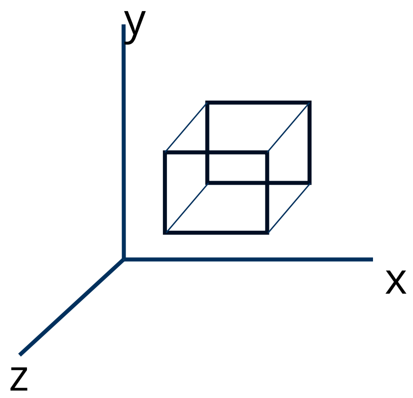


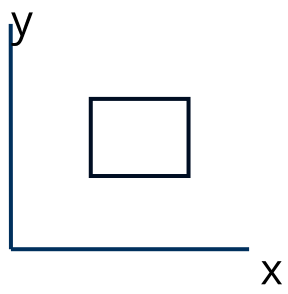
2D

Motivation

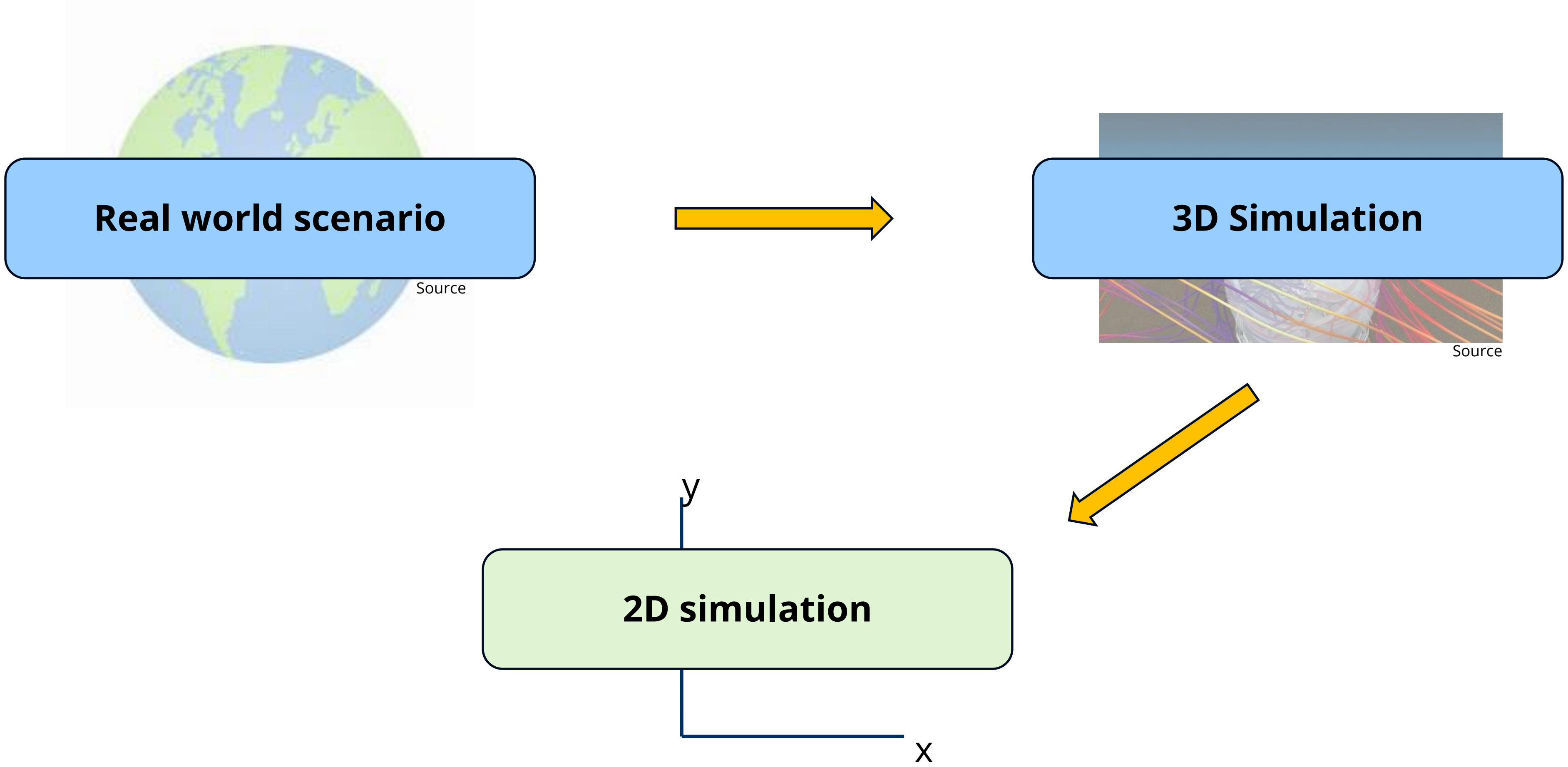
- Model simplification
- Dimensionality effects
- Computational cost
  - Lower cells
  - Faster meshing
  - Faster simulation

Disclaimer: A 2D model cannot depict reality !





Motivation



Real world scenario
Source
3D Simulation
Source
y
2D simulation
x

Contents

Introduction

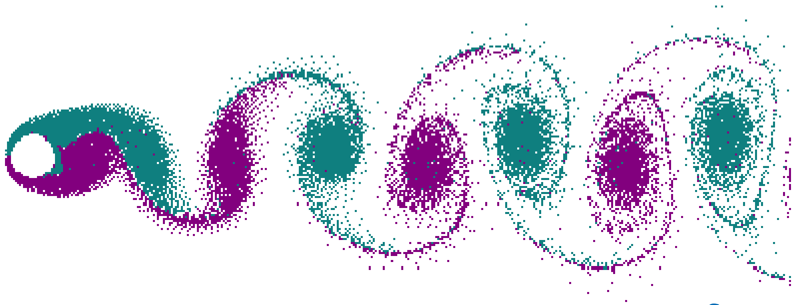
Background

- Kármán vortex street is a repeating pattern of swirling vortices, caused by a process known as vortex shedding.

- Vortex street forms only at a certain range of flow velocities characterised by the Reynolds number.

- Analysed using Strouhal number.

$$St = 0.198(1 - \frac{19.7}{Re})$$



Source

Source

Methodology

Simulation

Pre-processing

Methodology

Simulation

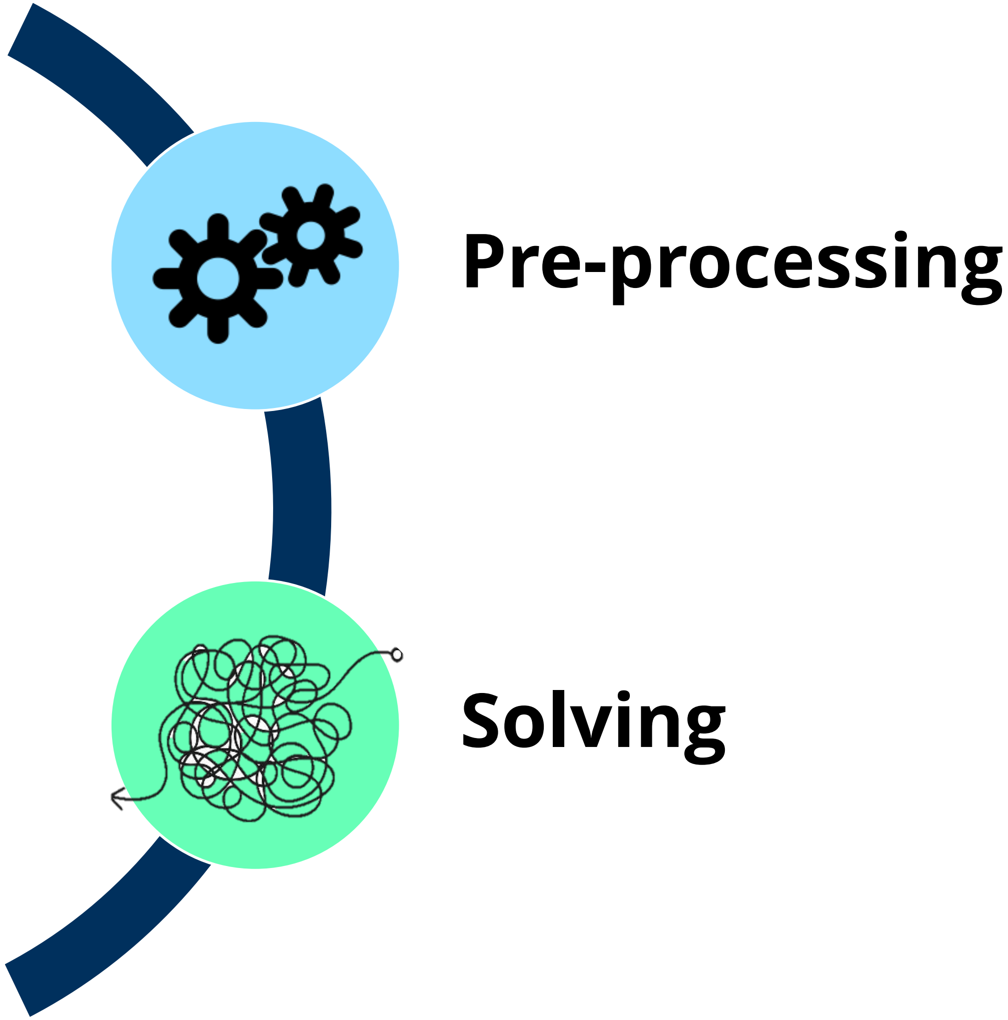
- Geometry Preparation
- Meshing
- Boundary Conditions
- Initial Conditions

Pre-processing

Methodology

Simulation

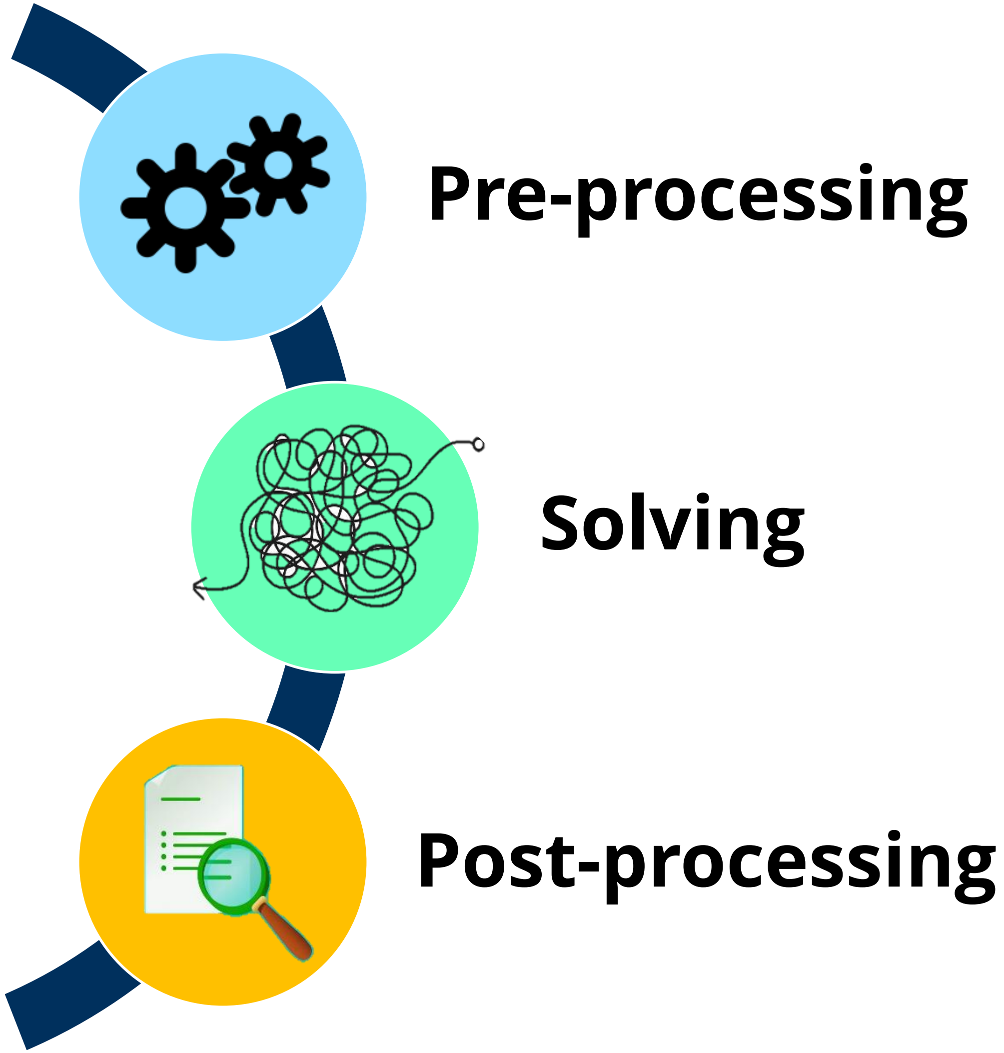
- Selection of Solver
- Turbulence Model
- Convergence Criteria
- Stopping Criteria



Methodology

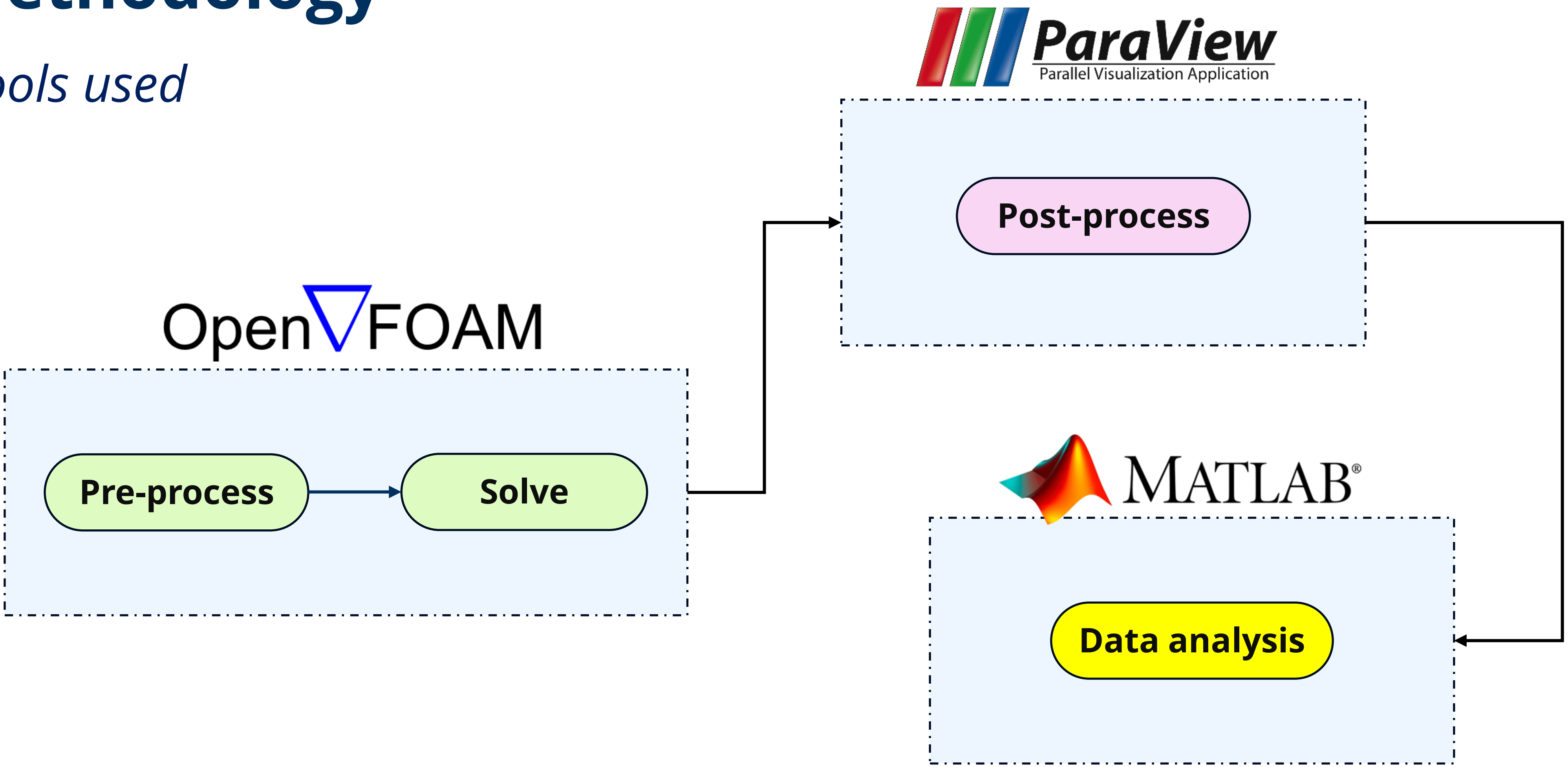
Simulation

- Data Extraction
- Data Visualization
- Quantitative Analysis
- Validation



Methodology

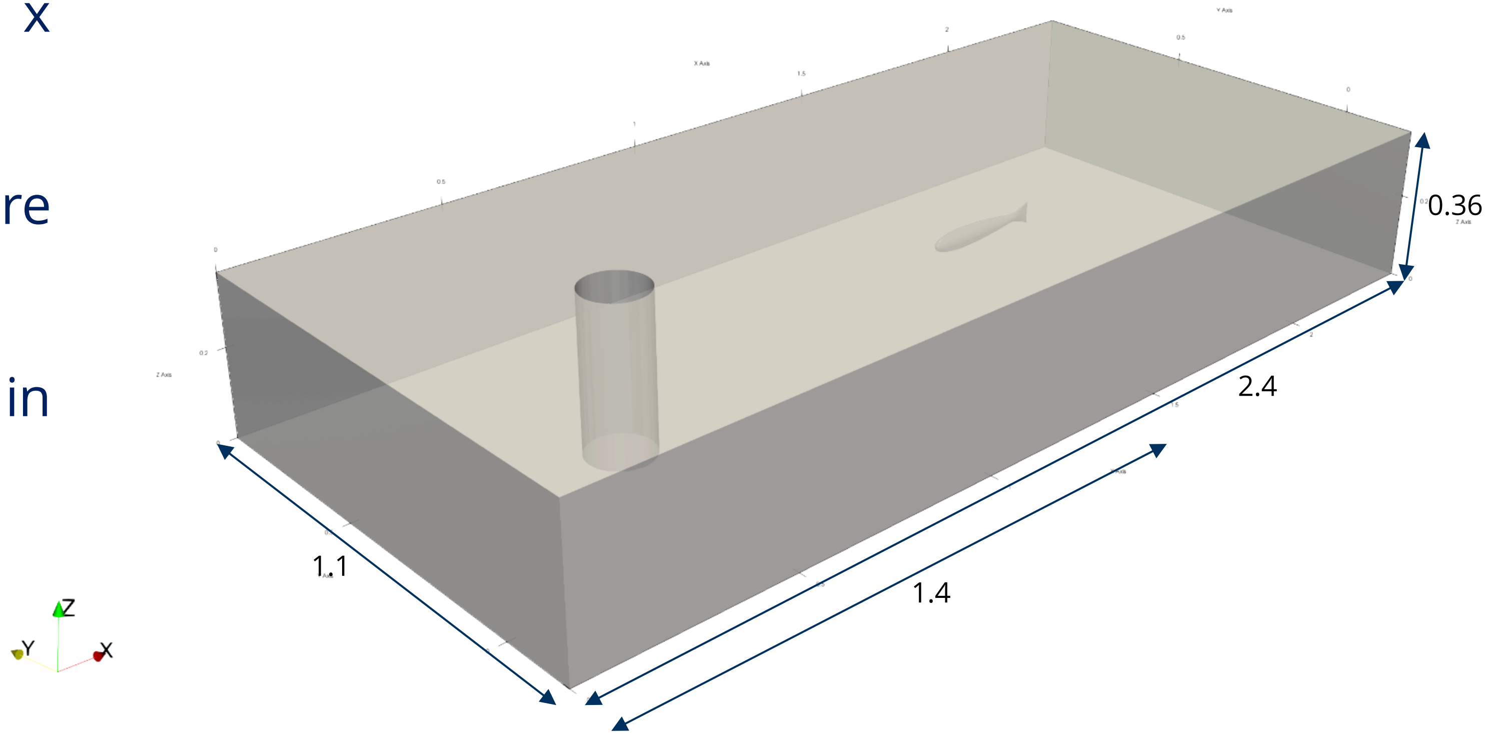
Tools used



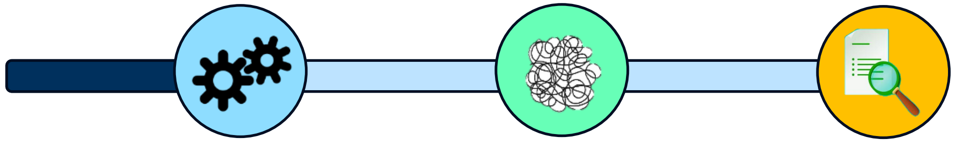
Methodology

Pre-process : Geometry

- Domain : 2.4m x 1.1m x 0.36m (l x b x h).
- Cylindrical structure (0.15m).
- Fish structure placed in vortex domain.

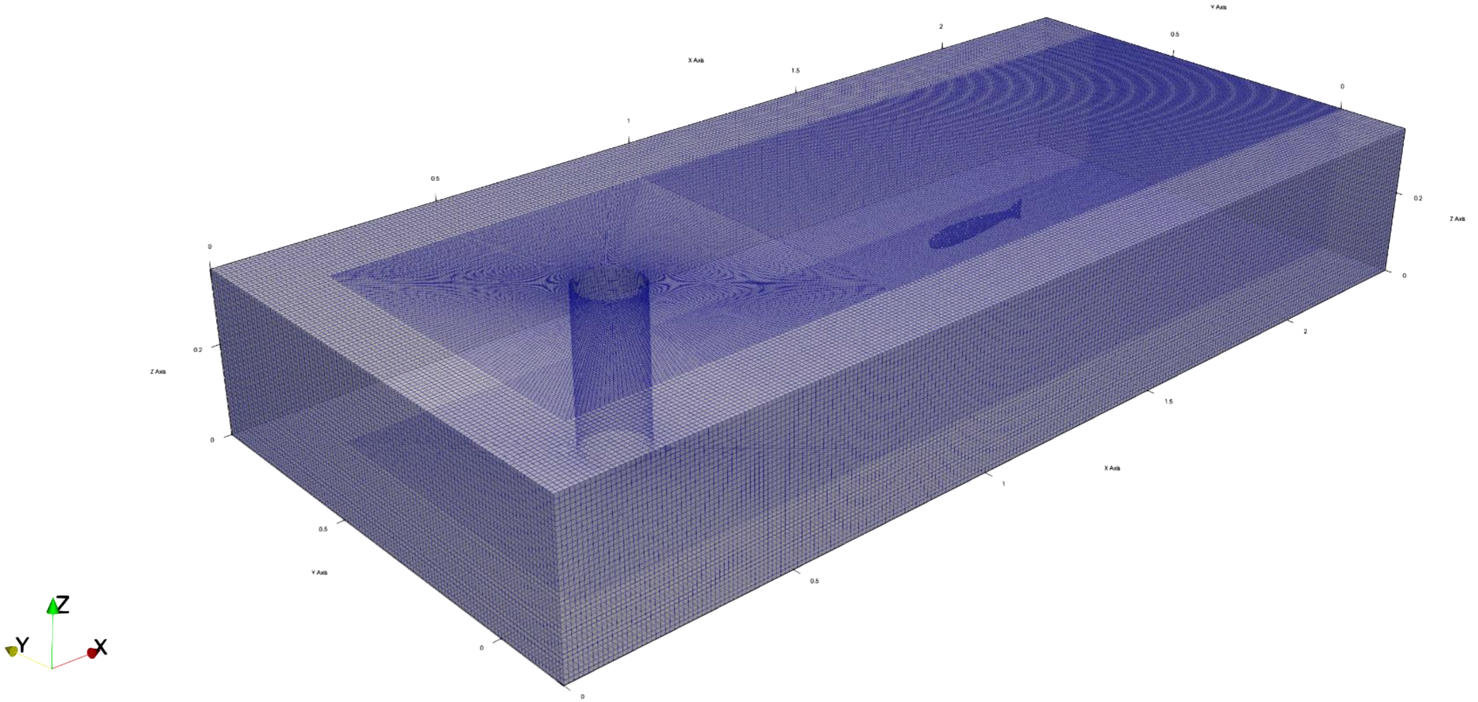


Methodology



Pre-process : Meshing

- Meshing using blockMesh and snappyHexMesh.
- blockMesh used to create structured and hexahedral meshes.
- snappyHexMesh used to generate high-quality, complex geometries meshes.



Methodology

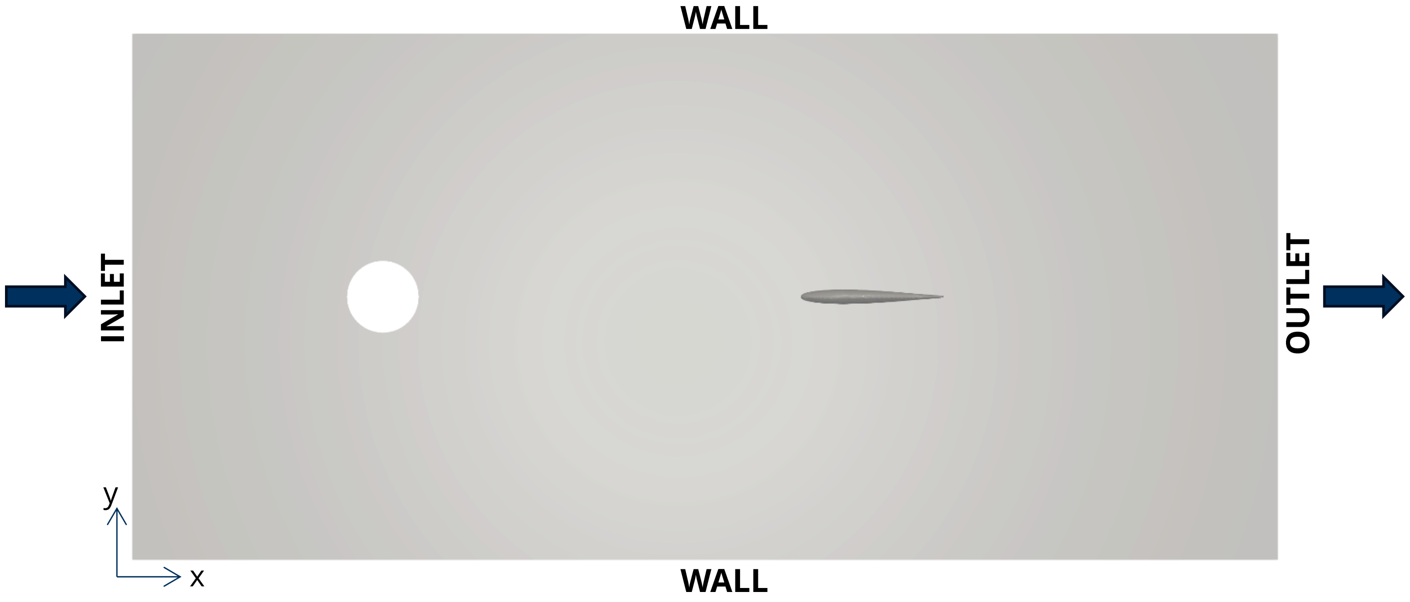
Pre-process : Initialisations

Boundary Conditions:

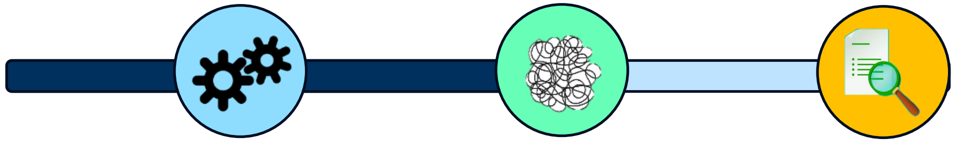
- Inlet: uniform velocity of 1m/s.
- Outlet: zero gradient pressure.
- Walls: slip
- Cylinder/ Fish: no slip

Governing equation: NS equation.

Turbulence model: *k*-*ω* SST.

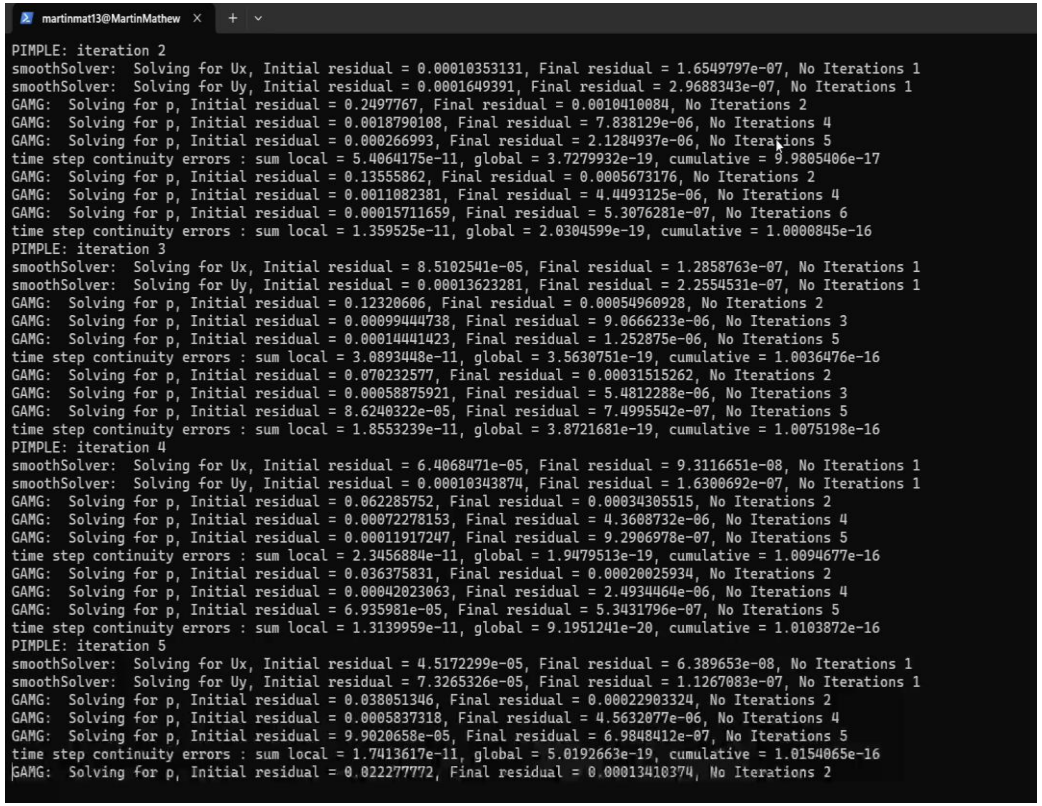


Methodology



Solve

- Navier-Stokes equation is solved until convergence is achieved.
- Convergence typically influenced by mesh resolution, time step-size and convergence criteria.
- PIMPLE solver: suitable for transient flows.



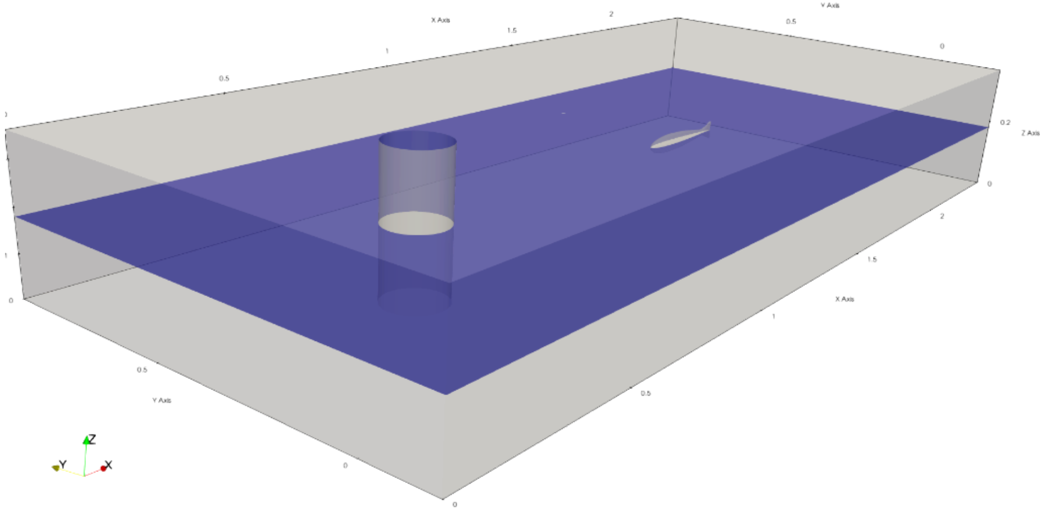
Results

Post-process

- Category: 1. qualitative / visualization based
  2. quantitative/ profile, trend based

- Post-process is the basis for comparison.

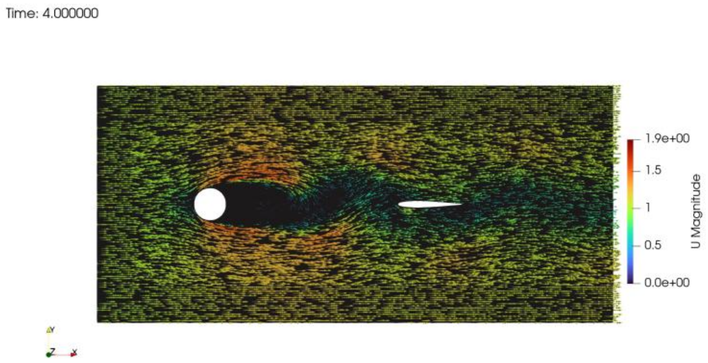
- Different post-process techniques employed at discrete times for quantitative analysis.



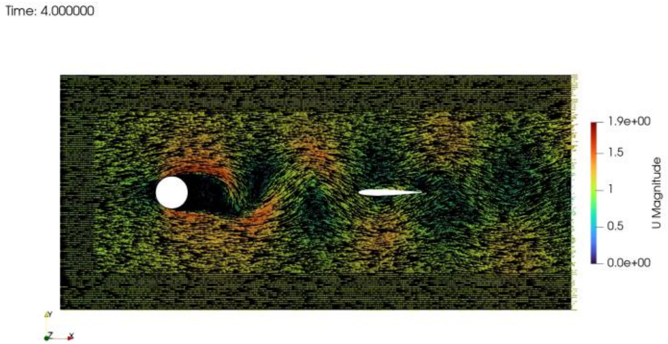
Results

Post-process : qualitative / visualization

- Visual Inspection of the system/ domain – Glyph.
- Domain is suitable for avoiding far-field effects.
- Far-field effects is the behaviour of fluid flow close to the boundaries.



2D

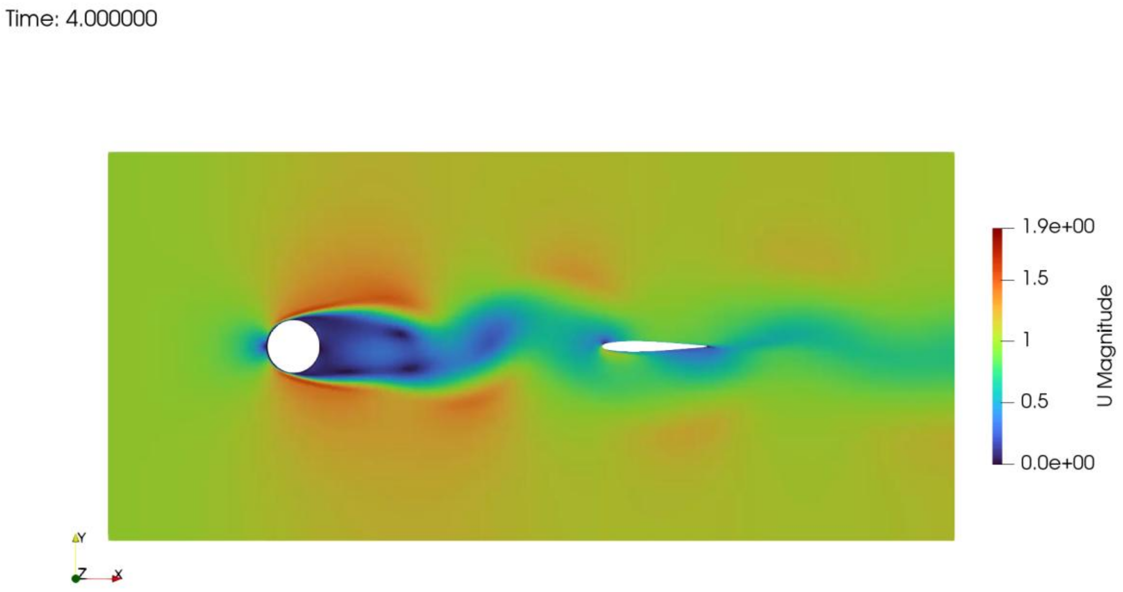


3D

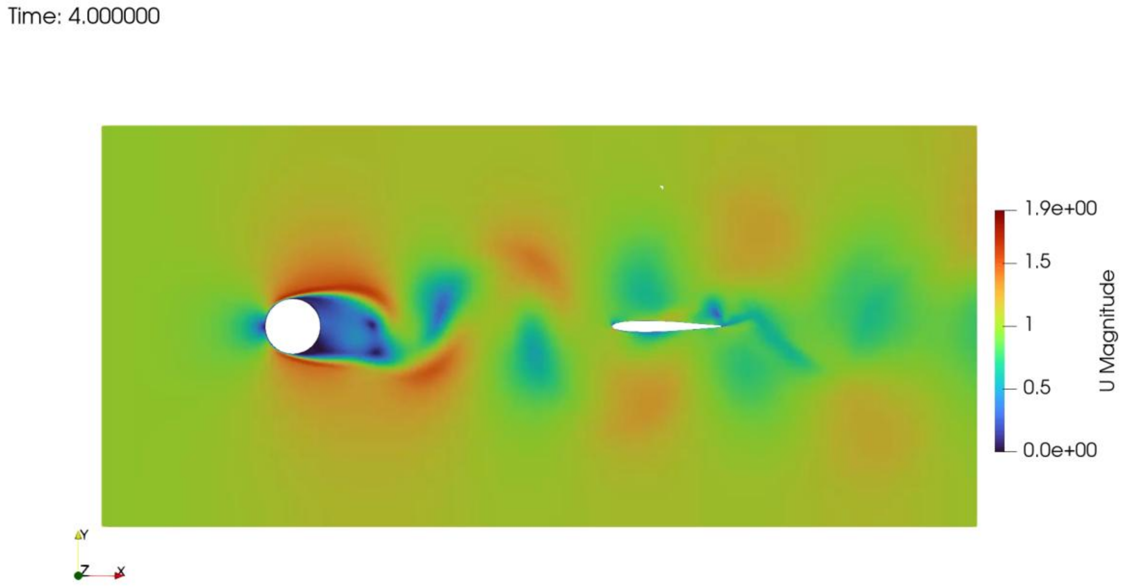
Results

Post-process : qualitative / visualization

- Visual Inspection of the system/ domain – Velocity pattern.



2D

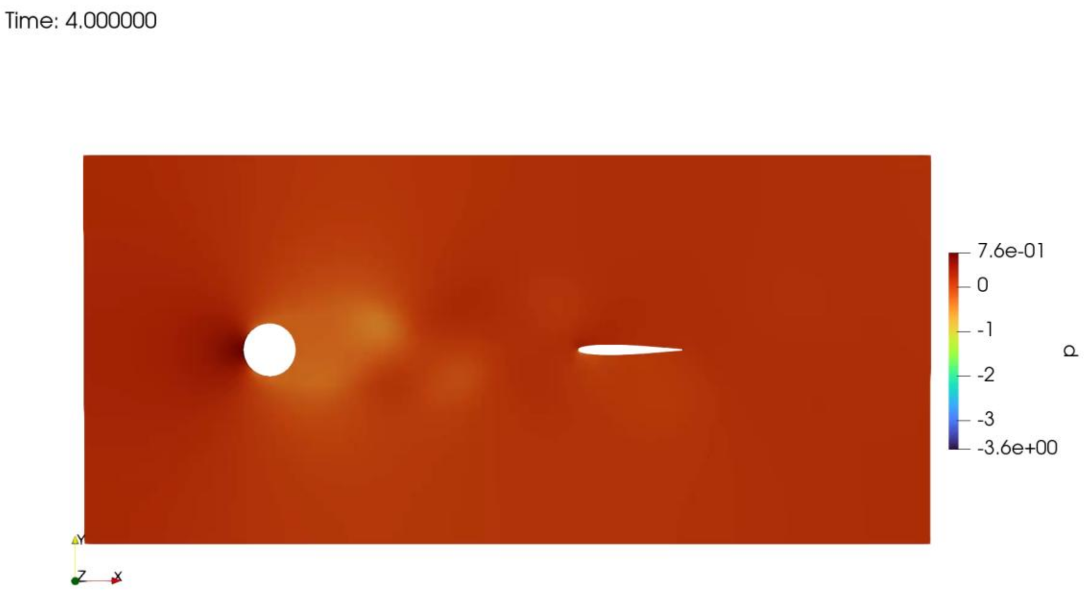


3D

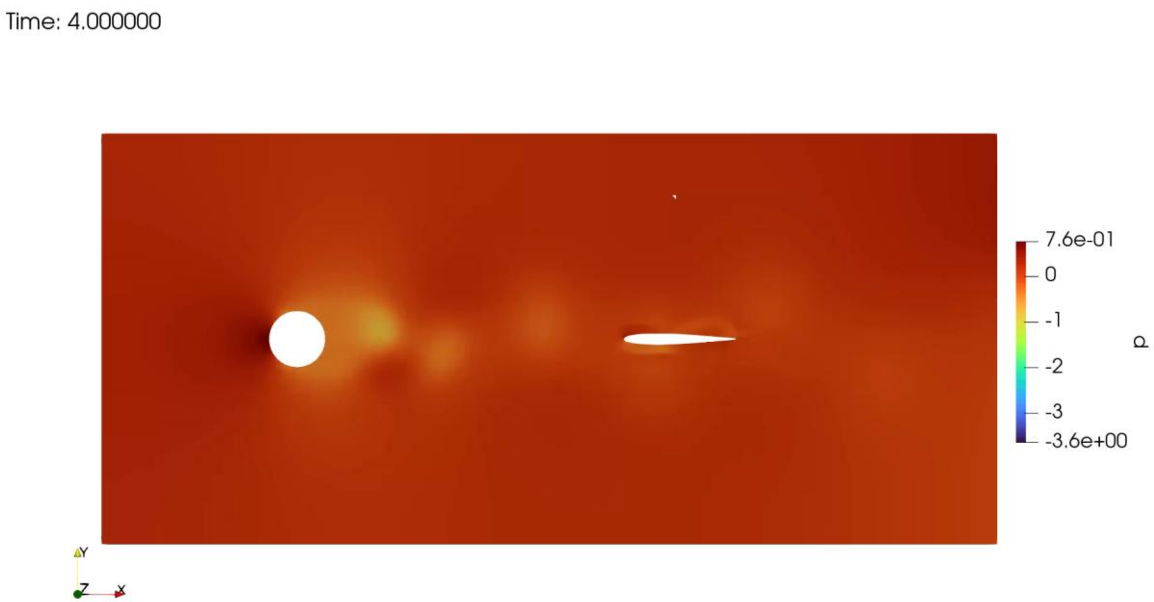
Results

Post-process : qualitative / visualization

- Visual Inspection of the system/ domain – Pressure pattern.



2D



3D

Results

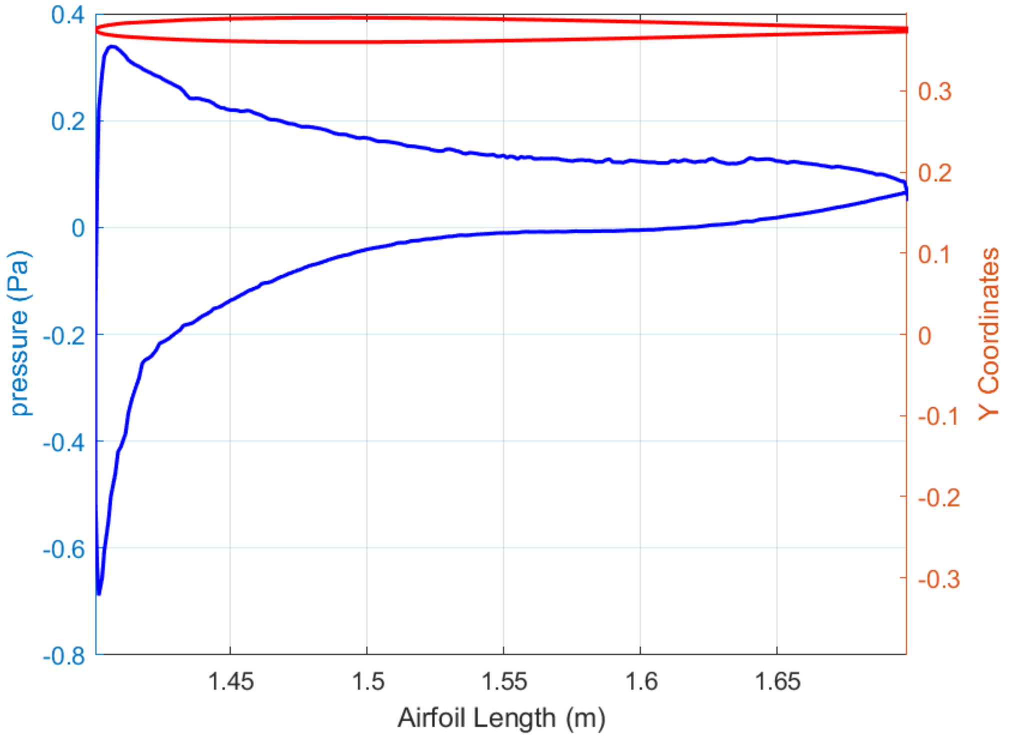
Post-process : quantitative analysis

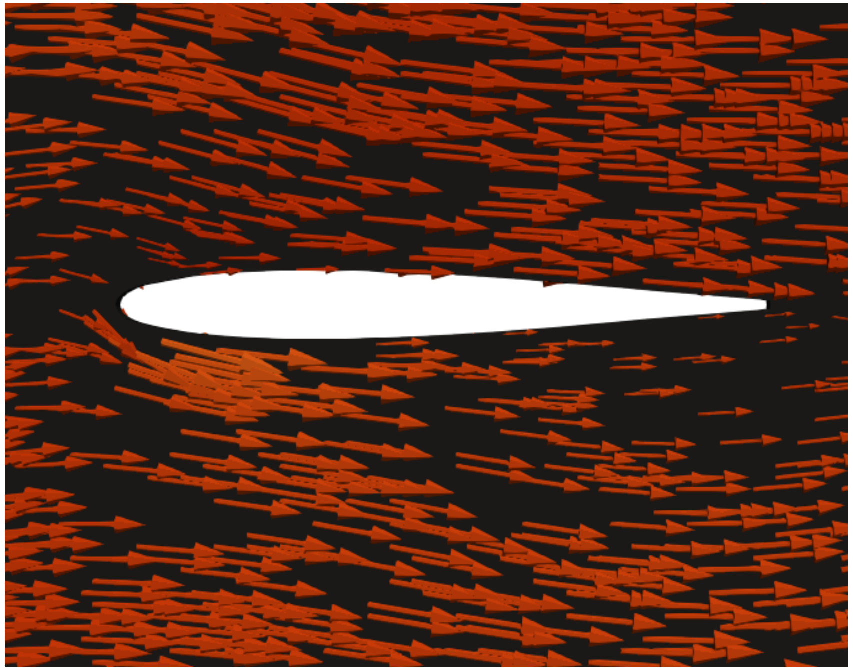
- Analysis of numerical data generated from the simulation.
- Plot over line functionality to quantify domain.
- Pressure distribution over the fish structure.
- Drag coefficient over the fish structure.

Results

Post-process : quantitative analysis

- Pressure variation at 4 second for 2D.

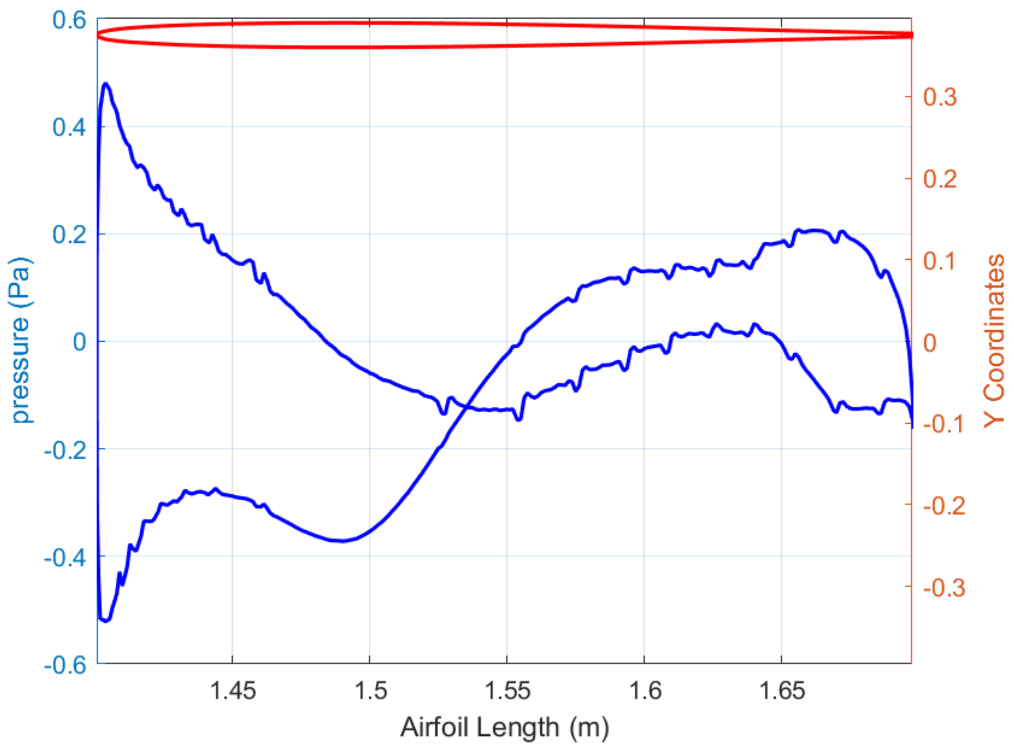


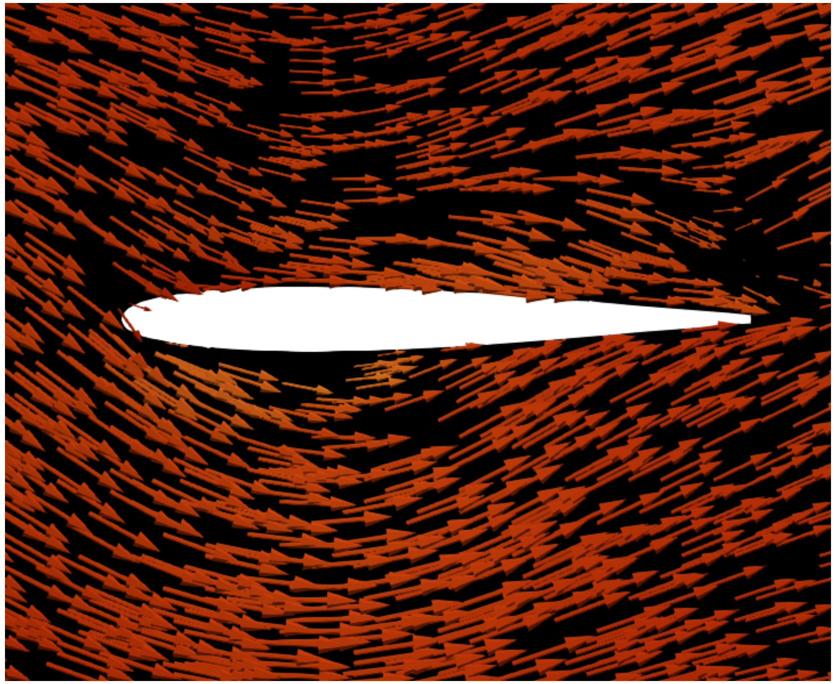


Results

Post-process : quantitative analysis

- Pressure variation at 4 second for 3D.





Conclusion

Inference

- While 2D model offers computational efficiency, 3D model provide a more realistic representation of complex flow phenomena.

- 2D model is suitable for qualitative analysis.

- 2D model may be implemented for dimension independent systems.

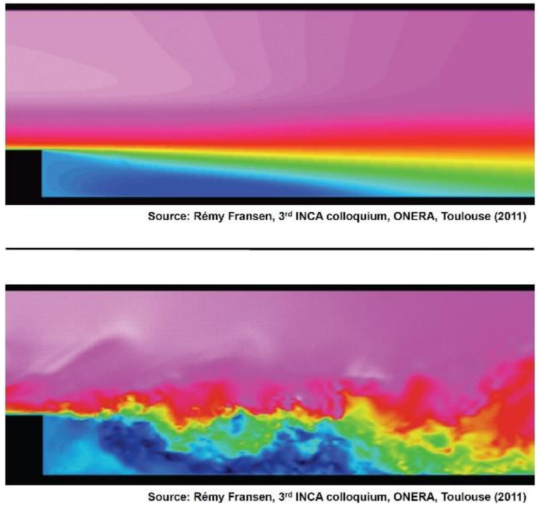
$$St = 0.198(1 - \frac{19.7}{Re})$$

- 3D model is suitable for quantitative analysis (pressure/ drag calculations).

Conclusion

Outlook

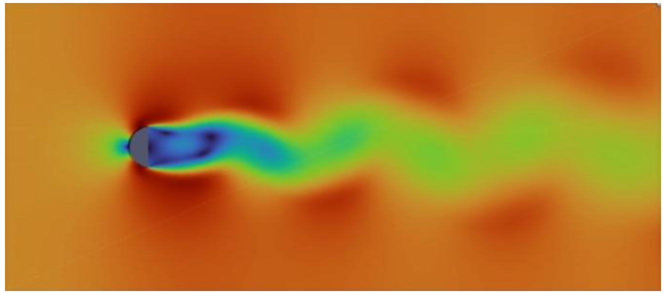
- Investigating the effects of different turbulence models (LES).
- Implementation in different numerical solver.
- Studying the impact of varying geometric parameters.
- Experimental validation.



RANS

LES

Source



Acknowledgment

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Thank you